Adrian M. Price-Whelan — Publication List

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Publications — *ADS search*

Refereed: 165 articles (18 first author)
Citations 61700
h-index: 60
(as of *2026-11-09*)

Refereed

165 Tavangar, K. *et al.* (incl. **APW**), *Phase Spirals across Galactic Disks. II. Using Large-scale "Macrospirals" in Phase-spiral Amplitude to Derive Perturbation Times*, ApJ, 1010, 126, 2026 (arXiv:2609.09543)

164 Arora, A. *et al.* (incl. **APW**), *No Stream Left Unscathed: The Imprint of a Host Galaxy*, ApJ, 1008, 91, 2026 (arXiv:2605.16200) [10 citations]

163 Holm, J. K. *et al.* (incl. **APW**), *Evidence for the first globular cluster stellar stream beyond the Milky Way*, Nature, 656, 843, 2026 (arXiv:2608.12254)

162 Rose, J. C. *et al.* (incl. **APW**), *The DREAMS Project: Disentangling the Impact of Halo-to-halo Variance and Baryonic Feedback on Milky Way Satellite Galaxies*, ApJ, 1006, 244, 2026 (arXiv:2512.02095) [4 citations]

161 Brooks, R. A. N. *et al.* (incl. **APW**), *The Milky Way–Large Magellanic Cloud interaction with simulation-based inference*, MNRAS, 550, 2026 (arXiv:2510.04735) [8 citations]

160 Brooks, R. A. N. *et al.* (incl. **APW**), *Quantifying the Milky Way, LMC and their interaction using all-sky kinematics of outer halo stars*, MNRAS, 549, 2026 (arXiv:2603.08788) [6 citations]

159 SDSS Collaboration *et al.* (incl. **APW**), *The Nineteenth Data Release of the Sloan Digital Sky Survey*, ApJS, 285, 9, 2026 (arXiv:2507.07093) [118 citations]

158 Sagear, S.; Ballard, S.; Daniel, K. J.; **Price-Whelan, A. M.** *et al.*, *The Orbital Eccentricities of Planets in the Kinematic Thin and Thick Galactic Disks*, AJ, 172, 42, 2026

157 Nibauer, J.; Bonaca, A.; **Price-Whelan, A. M.**; Spergel, D. N. *et al.*, *Measurement of Substructure from the Kinematics of the GD-1 Stellar Stream*, ApJ, 1004, 62, 2026 (arXiv:2510.02247) [33 citations]

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155 Lucey, M.; Mateu, C.; **Price-Whelan, A. M.**; Hogg, D. W. *et al.*, *Inferring Stellar Densities with Flexible Models. I. The Distribution of RR Lyrae in the Milky Way with Gaia DR3*, AJ, 171, 249, 2026 (arXiv:2510.03221) [5 citations]

154 Horta, D.; **Price-Whelan, A. M.**; Koposov, S. E.; Hunt, J. A. S. *et al.*, *The Milky Way's Circular Velocity Curve Measured Using Element Abundance Gradients*, ApJ, 1000, 8, 2026 (arXiv:2601.18876) [4 citations]

153 Hughes, J. *et al.* (incl. **APW**), *The Milky Way Bulge Extratidal Star Survey: NGC 6569*, AJ, 171, 137, 2026 (arXiv:2512.19074) [2 citations]

152 Otto, J. M. *et al.* (incl. **APW**), *The Open Cluster Chemical Abundances and Mapping Survey. VIII. Galactic Chemical Gradient and Azimuthal Analysis from SDSS/MWM DR19*, AJ, 171, 91, 2026 (arXiv:2507.07264) [32 citations]

151 Brooks, R. A. N. *et al.* (incl. **APW**), *A simulation-based inference approach to the dynamics of the MW–LMC system - validation*, MNRAS, 545, 2026 (arXiv:2507.10667) [17 citations]

150 Kollmeier, J. A. *et al.* (incl. **APW**), *Sloan Digital Sky Survey. V. Pioneering Panoptic Spectroscopy*, AJ, 171, 52, 2026 (arXiv:2507.06989) [157 citations]

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148 Chen, Y.; Gnedin, O. Y.; **Price-Whelan, A. M.**; Holm-Hansen, C., *StarStream: Automatic Detection Algorithm for Stellar Streams*, ApJ, 995, 15, 2025 (arXiv:2510.14929) [6 citations]

147 Viswanathan, A.; Horta, D.; **Price-Whelan, A. M.**; Starkenburg, E., *A slow spin to win: The gradual kinematic evolution across metallicities of the proto-Galaxy to the high-α disc*, A&A, 703, 2025 (arXiv:2411.12165) [16 citations]

146 Oeur, M.; Loebman, S. R.; **Price-Whelan, A. M.**; Arora, A. *et al.*, *OTI on FIRE: Testing the Efficacy of Orbital Torus Imaging to Recover the Galactic Potential*, ApJ, 993, 29, 2025 (arXiv:2505.05590) [2 citations]

145 Bernet, M.; Ramos, P.; Antoja, T.; **Price-Whelan, A. M.** *et al.*, *Dynamics of tidal spiral arms: Machine learning-assisted identification of equations and application to the Milky Way*, A&A, 702, 2025 (arXiv:2506.17383) [7 citations]

144 Chance, Q. *et al.* (incl. **APW**), *paired: A Statistical Framework for Detecting Stellar Binarity with Gaia RVs. I. Sensitivity to Unresolved Binaries*, ApJ, 992, 131, 2025 (arXiv:2206.11275) [16 citations]

143 Adams, D. K. *et al.* (incl. **APW**), *Semianalytic Modeling of Dark Matter Subhalo Encounters with Thin Stellar Streams: Statistical Predictions for GD-1-like Streams in Cold Dark Matter*, ApJ, 991, 66, 2025 (arXiv:2412.13144) [11 citations]

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141 Wagg, T. *et al.* (incl. **APW**), *Delayed and Displaced: The Impact of Binary Interactions on Core-collapse SN Feedback*, AJ, 170, 192, 2025 (arXiv:2504.17903) [14 citations]

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139 Filion, C. *et al.* (incl. **APW**), *Counterculture Stars: Slow and Retrograde Stars with Low-alpha Disk Abundances*, ApJ, 989, 70, 2025 (arXiv:2506.09927)

138 Tavangar, K.; **Price-Whelan, A. M.**, *Inferring the Density and Membership of Stellar Streams*

with Flexible Models: The GD-1 Stream in Gaia Data Release 3, ApJ, 988, 45, 2025 (arXiv:2502.13236) [14 citations]

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132 Nibauer, J.; Bonaca, A.; Spergel, D. N.; **Price-Whelan, A. M.** *et al.*, *StreamSculptor: Hamiltonian Perturbation Theory for Stellar Streams in Flexible Potentials with Differentiable Simulations*, ApJ, 983, 68, 2025 (arXiv:2410.21174) [26 citations]

131 Starkman, N.; **Price-Whelan, A. M.**; Nibauer, J., *unxt: A Python package for unit-aware computing with JAX*, JOSS, 10, 7771, 2025 (arXiv:2603.08770) [3 citations]

130 Lu, Y. *et al.* (incl. **APW**), *Evidence of Truly Young High-α Dwarf Stars*, AJ, 169, 168, 2025 (arXiv:2410.02962) [12 citations]

129 Starkman, N. *et al.* (incl. **APW**), *Stream Members Only: Data-driven Characterization of Stellar Streams with Mixture Density Networks*, ApJ, 980, 253, 2025 (arXiv:2311.16960) [14 citations]

128 **Price-Whelan, A. M.**; Hunt, J. A. S.; Horta, D.; Oeur, M. *et al.*, *Data-driven Dynamics with Orbital Torus Imaging: A Flexible Model of the Vertical Phase Space of the Galaxy*, ApJ, 979, 115, 2025 (arXiv:2401.07903) [10 citations]

127 Foote, H. R. *et al.* (incl. **APW**), *Segue 2 Recently Collided with the Cetus-Palca Stream: New Opportunities to Constrain Dark Matter in an Ultra-faint Dwarf*, ApJ, 979, 171, 2025 (arXiv:2408.06415) [5 citations]

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125 Wagg, T.; Breivik, K.; Renzo, M.; **Price-Whelan, A. M.**, *cogsworth: A Gala of COSMIC Proportions Combining Binary Stellar Evolution and Galactic Dynamics*, ApJS, 276, 16, 2025 (arXiv:2409.04543) [19 citations]

124 Brooks, R. A. N.; Garavito-Camargo, N.; Johnston, K. V.; **Price-Whelan, A. M.** *et al.*, *LMC Calls, Milky Way Halo Answers: Disentangling the Effects of the MW–LMC Interaction on Stellar Stream Populations*, ApJ, 978, 79, 2025 (arXiv:2410.02574) [26 citations]

123 Hackshaw, Z. *et al.* (incl. **APW**), *[X/Fe] Marks the Spot: Mapping Chemical Azimuthal Varia-*

tions in the Galactic Disk with APOGEE, ApJ, 977, 143, 2024 (arXiv:2405.18120) [29 citations]

122 Sagear, S.; **Price-Whelan, A. M.**; Ballard, S.; Lu, Y. (. *et al.*, *zoomies: A Tool to Infer Stellar Age from Vertical Action in Gaia Data*, ApJ, 977, 49, 2024 (arXiv:2403.09878) [8 citations]

121 Garavito-Camargo, N.; **Price-Whelan, A. M.**; Samuel, J.; Cunningham, E. C. *et al.*, *On the Corotation of Milky Way Satellites: LMC-mass Satellites Induce Apparent Motions in Outer Halo Tracers*, ApJ, 975, 100, 2024 (arXiv:2311.11359) [20 citations]

120 Foster, S. *et al.* (incl. **APW**), *Carbon enrichment in APOGEE disk stars as evidence of mass transfer in binaries*, A&A, 689, 2024 (arXiv:2407.18130) [6 citations]

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118 Castro-Ginard, A. *et al.* (incl. **APW**), *Gaia DR3 detectability of unresolved binary systems*, A&A, 688, 2024 (arXiv:2404.14127) [130 citations]

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116 Cunningham, E. C.; Hunt, J. A. S.; **Price-Whelan, A. M.**; Johnston, K. V. *et al.*, *Chemical Cartography of the Sagittarius Stream with Gaia*, ApJ, 963, 95, 2024 (arXiv:2307.08730) [35 citations]

115 Hunt, J. A. S.; **Price-Whelan, A. M.**; Johnston, K. V.; McClure, R. L. *et al.*, *Radial phase spirals in the Solar neighbourhood*, MNRAS, 527, 11393, 2024 (arXiv:2401.08748) [17 citations]

114 Horta, D.; **Price-Whelan, A. M.**; Hogg, D. W.; Johnston, K. V. *et al.*, *Orbital Torus Imaging: Acceleration, Density, and Dark Matter in the Galactic Disk Measured with Element Abundance Gradients*, ApJ, 962, 165, 2024 (arXiv:2312.07664) [28 citations]

113 Aganze, C. *et al.* (incl. **APW**), *Prospects for Detecting Gaps in Globular Cluster Stellar Streams in External Galaxies with the Nancy Grace Roman Space Telescope*, ApJ, 962, 151, 2024 (arXiv:2305.12045) [17 citations]

112 Darragh-Ford, E.; Hunt, J. A. S.; **Price-Whelan, A. M.**; Johnston, K. V., *ESCARGOT: Mapping Vertical Phase Spiral Characteristics Throughout the Real and Simulated Milky Way*, ApJ, 955, 74, 2023 (arXiv:2302.09086) [38 citations]

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108 Silva, É. *et al.* (incl. **APW**), *Extratidal Members of Segue 3 are Rare and Difficult to Confirm*, RNAAS, 7, 127, 2023

107 Hawkins, K.; **Price-Whelan, A. M.**; Sheffield, A. A.; Subrahimovic, A. Z. *et al.*, *On the Hunt for the Origins of the Orphan-Chenab Stream: Detailed Element Abundances with APOGEE and*

Gaia, ApJ, 948, 123, 2023 (arXiv:2205.14218) [15 citations]

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104 Chamberlain, K.; **Price-Whelan, A. M.**; Besla, G.; Cunningham, E. C. *et al.*, *Implications of the Milky Way Travel Velocity for Dynamical Mass Estimates of the Local Group*, ApJ, 942, 18, 2023 (arXiv:2204.07173) [34 citations]

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96 Astropy Collaboration; **Price-Whelan, A. M.**; Lim, P. L.; Earl, N. *et al.*, *The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package*, ApJ, 935, 167, 2022 (arXiv:2206.14220) [6400 citations]

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83 Kounkel, M.; Covey, K. R.; Stassun, K. G.; **Price-Whelan, A. M.** *et al.*, *Double-lined Spectroscopic Binaries in the APOGEE DR16 and DR17 Data*, AJ, 162, 184, 2021 (arXiv:2107.10860) [89 citations]

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66 **Price-Whelan, A. M.**; Hogg, D. W.; Rix, H.; Beaton, R. L. *et al.*, *Close Binary Companions to APOGEE DR16 Stars: 20,000 Binary-star Systems Across the Color-Magnitude Diagram*, ApJ, 895, 2, 2020 (arXiv:2002.00014) [135 citations]

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